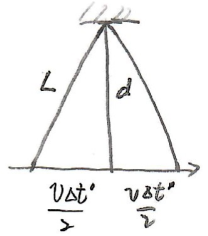
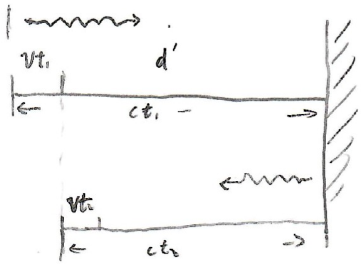
11.4 (a) When the clock is at rest, the time interval between sending and receiving a light signal is $\Delta t = 2d/c$. When the clock moves, from the observer's perspective, the signal will traverse longer distance, and the time interval is



$$\Delta t' = 2L/c = \frac{2}{c}\sqrt{\left(\frac{c\Delta t}{2}\right)^2 + \left(\frac{v\Delta t'}{2}\right)^2}, \quad \text{or} \quad (\Delta t')^2 = (\Delta t)^2 + \left(\frac{v}{c}\right)^2 (\Delta t')^2.$$

Then $\Delta t' = \dfrac{\Delta t}{\sqrt{1 - v^2/c^2}}$. Since $\Delta t' > \Delta t$, for ~~the~~ observer, the clock now ticks slower.

(b) Consider the configuration on the right, we have



$$d' + vt_1 = ct_1$$

$$d' - vt_2 = ct_2$$

and $\Delta t' = t_1 + t_2 = \dfrac{d'}{c-v} + \dfrac{d'}{c+v} = \dfrac{d' \cdot 2c}{c^2 - v^2} = \dfrac{2d'}{c} \cdot \dfrac{1}{1 - v^2/c^2}$

From the observer's pespective, $d' = d\sqrt{1 - v^2/c^2}$ by Lorentz contraction. Then

$$\Delta t' = \frac{2d}{c}\,\frac{1}{\sqrt{1 - v^2/c^2}} = \frac{\Delta t}{\sqrt{1 - v^2/c^2}}$$